

Design Brief **Template**

Student Name: [Your Name]

Team Name: [Hour/Group #]

Date: [Submission Date]

Instructor: Mr. Bombich

Project Overview: This design brief outlines the initial stages of the Engineering Design Process (EDP) for an engineering project. It covers the Ask, Research, and Imagine steps, with an evaluation of solutions using a Pugh Matrix to guide the Plan step. The project addresses [Summarize Project].

1. Problem Statement (Ask Step)

Criteria per Rubric: Clearly define the project problem, identifying specific stakeholders, needs, and constraints (e.g., budget, materials, time). Thoroughly integrate ethical considerations (e.g., sustainability, safety). Include insightful questions for the “customer” (instructor or stakeholder) tied to project goals to achieve Exceeds Expectations (90–100%).

Content: Describe the problem, including who is affected, what the issue is, where and when it occurs, why it matters, and how much impact it has. List stakeholders (e.g., users, organizations), needs (e.g., functionality, ease of use), and constraints (e.g., cost, size, timeline). Address ethical considerations (e.g., environmental or societal impact). Provide 2–3 questions you would ask the “customer” to clarify requirements.

Example Placeholder:

The problem is [describe issue, e.g., inefficiency in a process]. Stakeholders include [list groups]. The needs are [list requirements]. The constraints include [budget, materials, time]. Ethical considerations involve [e.g., sustainability]. Questions for the customer were: [list 2–3 specific questions and answers].

2. Research Summary (Research Step)

Criteria per Rubric: Summarize 3–5 credible sources (e.g., technical guides, industry articles) with detailed insights relevant to the project’s goals and constraints. Clearly connect research to project needs and ethical considerations (e.g., environmental impact). Provide complete citations, including AI tool use if applicable (per Lab Rules Section 9), to achieve Exceeds Expectations (90–100%).

Content: Summarize findings from 3–5 credible sources (e.g., Arduino.cc, industry websites, or academic articles) related to existing solutions, components, or technologies relevant to the project. Explain how research addresses project constraints (e.g., cost, materials) and ethical considerations (e.g., energy efficiency). Include full citations (source, date, and AI tool details if

used, e.g., “Generated with Grok by xAI, [date]”).

Example Placeholder:

Research includes [summarize 3–5 sources, e.g., technical guide on sensors]. Findings show [describe solutions/components]. This addresses constraints like [e.g., cost] and ethics like [e.g., sustainability]. Citations: [list sources with dates; include AI use if applicable].

3. Brainstormed Ideas (Imagine Step)

Criteria per Rubric: Present 5+ creative, diverse ideas with detailed descriptions and clear, labeled paper/pencil sketches (photos uploaded or described). Ensure ideas address project constraints and ethical considerations. Demonstrate team collaboration through detailed journal entries documenting individual roles to achieve Exceeds Expectations (90–100%).

Content: List 5+ distinct ideas for solving the problem, each with a brief description (2–3 sentences) and a description of a paper/pencil sketch (or upload photos to Google Docs). Explain how each idea addresses constraints and ethical considerations. Document team collaboration in journal entries (e.g., who sketched, who brainstormed).

Example Placeholder:

- Idea 1: [Describe solution]. Pros: [list benefits]. Cons: [list drawbacks]. Addresses [constraints/ethics]. Sketch: [describe sketch, e.g., labeled diagram of mechanism].
- [Repeat for 3-5 ideas]. Collaboration: [summarize team roles, supported by journals].

4. Evaluation (Pugh Matrix)

Criteria per Rubric: Provide a Pugh Matrix comparing all ideas to a baseline using 3–5 weighted criteria (e.g., functionality, cost, feasibility, ethics). Ensure math is accurate (scores: +1, 0, -1 × weights, summed correctly). Clearly justify the top solution, tying it to project goals and ethical considerations, to achieve Exceeds Expectations (90–100%).

Content: Create a Pugh Matrix table comparing all ideas to a chosen baseline (e.g., Idea 1). List 3–5 criteria with weights (1–5 scale, higher = more important). Score each idea relative to the baseline (+1 for better, 0 for same, -1 for worse). Calculate weighted scores (score × weight) and sum for totals. Justify the top-scoring solution, explaining how it meets project goals and ethics.

Example Placeholder:

Criteria	Weight	Baseline (Idea 1)	Idea 2	Idea 3	Idea 4	Idea 5
[Criterion, e.g., Functionality]	[e.g., 5]	S	[score]	[score]	[score]	[score]
[Repeat for 3–5 criteria]						

Total Score

0

[sum]

[sum]

[sum]

[sum]

Selected Solution: [Top idea] because [justify with project goals/ethics].